

Portfolio & Risk Intelligence Report

Exposures, risk, stress tests and limit monitoring

Prepared for	Demo Client
Engagement	DEMO-0002
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Executive summary

Demonstration report. Built entirely from synthetic data generated by ProQuant for illustration. No real securities or client data are used.

ProQuant Analytics analysed the holdings of **Demo Multi-Asset Fund** as of **2025-12-31**: exposures and concentration, volatility and risk contributions, factor exposure, Value-at-Risk and Expected Shortfall, stress tests, and compliance with the risk limits agreed with Demo Client.

Risk status: Limit breach - 1 limit(s) exceeded, action required

Measured against the limits in the client's risk policy (table below). Risk estimates are model-based and describe the current portfolio; they are not forecasts of returns.

NET ASSET VALUE	EX-ANTE VOLATILITY	VAR 99% 1-DAY
USD 362.3m	**13.6%**	**2.01%**
39 positions	annualised, EWMA	USD 7.3m

EXPECTED SHORTFALL 97.5%	WORST STRESS LOSS	EQUITY BETA
2.16%	**22.6%**	**0.70**
1-day, historical	COVID-19 crash	to market factor

Key findings

1. As of 2025-12-31, the portfolio holds 39 positions with a NAV of USD 362.3m; the largest position is Synthetic Equity 01 at 8.9% and the top 10 make up 49.1% (effective number of positions 24.7).
2. Ex-ante volatility is 13.6% a year (EWMA; 12.6% with equal weighting). Diversification reduces stand-alone risk by a factor of 1.66.
3. One-day 99% VaR is 2.01% of NAV (USD 7.3m) by historical simulation; parametric and Monte Carlo estimates are 2.00% and 2.25%. 97.5% Expected Shortfall is 2.16%.
4. The VaR model backtest shows 5 exceptions in 500 days against 5.0 expected (Basel zone GREEN; Kupiec p-value 1.00).
5. Factor model R^2 0.93: equity market beta 0.70; 79.4% of variance is explained by the factors.
6. The worst stress scenario is 'COVID-19 crash' with a loss of 22.6% (-USD 81.7m).
7. Limit breaches: Largest single position (% NAV).
8. On watch (above 90% of limit): Top-10 positions (% NAV); Ex-ante volatility (annual).
9. 1.2% of NAV relies partly on proxy return histories (short track records).

Limit monitoring

Limit	Measured	Limit	Utilisation	Status
Largest single position (% NAV)	8.9%	<= 8.0%	111%	BREACH
Top-10 positions (% NAV)	49.1%	<= 50.0%	98%	WATCH
Largest sector (% NAV)	17.9%	<= 30.0%	60%	OK
Effective number of positions	24.73	>= 15.00	61%	OK
Ex-ante volatility (annual)	13.6%	<= 15.0%	91%	WATCH
1-day 99% VaR, historical (% NAV)	2.0%	<= 2.5%	81%	OK
1-day 97.5% Expected Shortfall (% NAV)	2.2%	<= 3.0%	72%	OK
Beta to equity market factor	0.70	<= 1.00	70%	OK
Worst stress-scenario loss (% NAV)	22.6%	<= 30.0%	75%	OK
Cash buffer (% NAV)	3.0%	>= 2.0%	67%	OK

Source: tables/limits.csv. WATCH = utilisation at or above 90% of the limit.

Recommended actions

- Reduce the largest position below the single-name limit or document an approved exception.
- Review concentration in the top-10 positions; new purchases should broaden the portfolio.
- Ex-ante volatility is close to or above limit; consider de-risking or hedging equity beta.

Methodology

This report tells where the risk sits and what to do about it. The steps, formulas and references are in the methodology document. One thing helps when reading on: every risk number describes the holdings of today, not the portfolio's past.

Results

1. Portfolio and performance

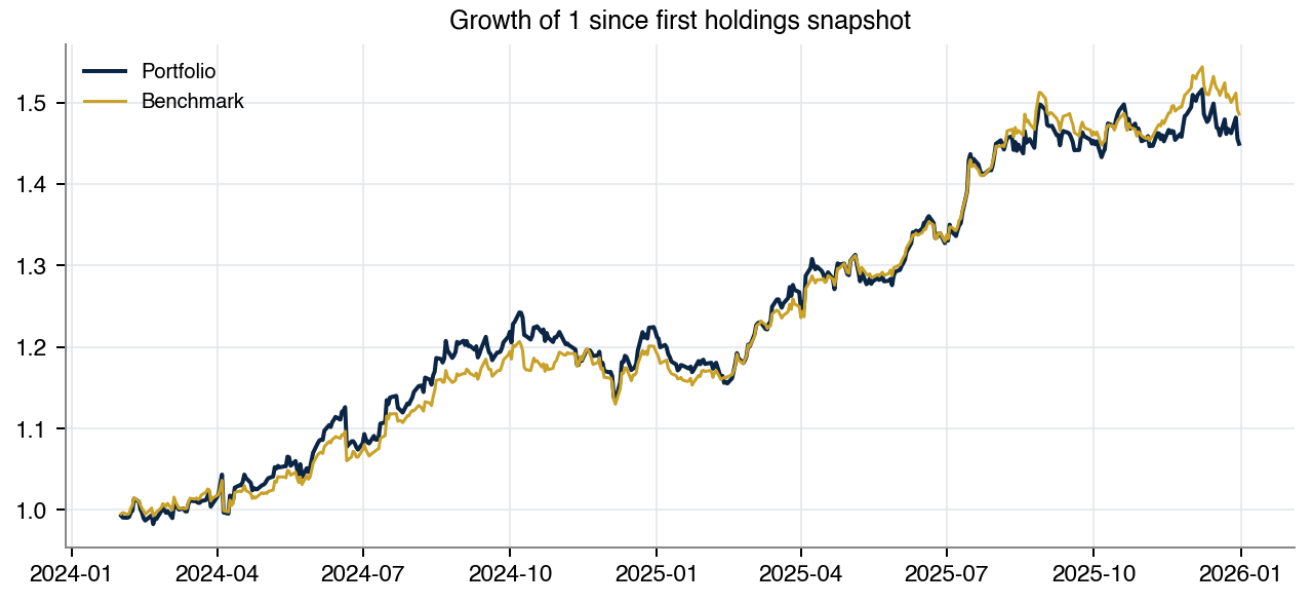


Figure 1. Realized performance from holdings. Source: tables/equity_curves.csv

Metric	Portfolio	Benchmark
Total return	44.9%	48.6%
Annualised return	20.6%	22.1%
Volatility	12.3%	9.9%
Sharpe ratio	1.58	2.06
Max drawdown	-8.7%	-6.4%
Worst day	-4.5%	-3.6%
Beta to benchmark	1.18	n/a
Tracking error	4.1%	n/a
Information ratio	-0.24	n/a

Source: tables/performance_summary.csv

Monthly returns (%)

year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2024		-0.4	1.1	2.3	3.8	0.3	5.6	6.1	0.7	0.2	-3.6	4.1	21.9
2025	-3.3	1.8	5.6	1.8	0.2	2.7	8.2	4.3	-3.2	0.2	2.2	-2.4	18.9

Source: tables/monthly_returns.csv (%)

2. Exposures and concentration

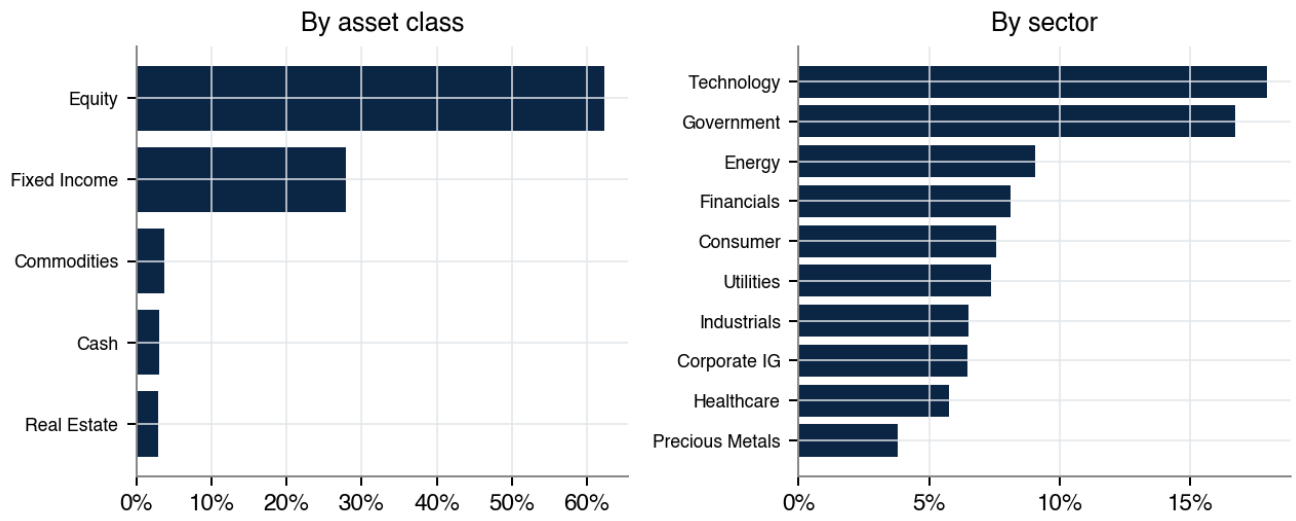


Figure 2. Exposure by asset class and sector. Source: tables/exposure_*.csv

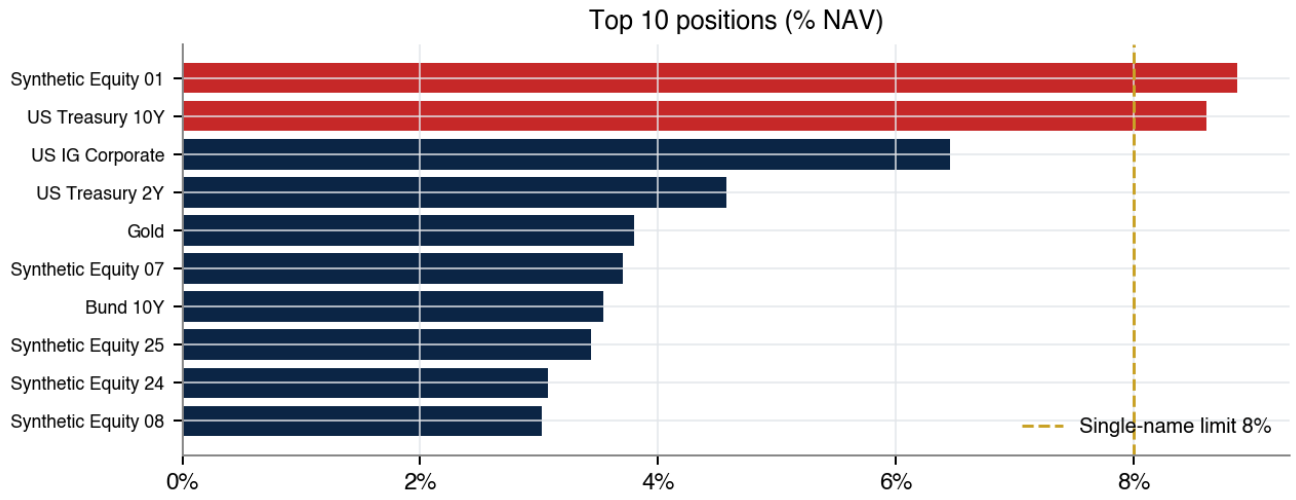


Figure 3. Largest positions (red = above single-name limit). Source: tables/positions.csv

Asset class	Market value	Weight	Positions
Equity	USD 225.9m	62.4%	31
Fixed Income	USD 101.0m	27.9%	6
Commodities	USD 13.8m	3.8%	1
Cash	USD 10.9m	3.0%	1
Real Estate	USD 10.7m	3.0%	1

Source: tables/exposure_asset_class.csv

Country	Weight
US	63.0%
DE	10.4%
CH	7.7%
GB	7.5%
JP	6.2%
FR	5.2%

Source: tables/exposure_country.csv

Currency	Weight
USD	100.0%

Source: tables/exposure_currency.csv

Concentration: top 10 = 49.1%, Herfindahl index 0.040, effective number of positions 24.7; largest sector Technology at 17.9%.

3. Volatility, correlation and risk contributions

Ex-ante volatility 13.6% (EWMA) / 12.6% (equal weights); realized 3-month volatility 12.0%. Diversification ratio 1.66; average pairwise correlation of the 15 largest positions 0.19.

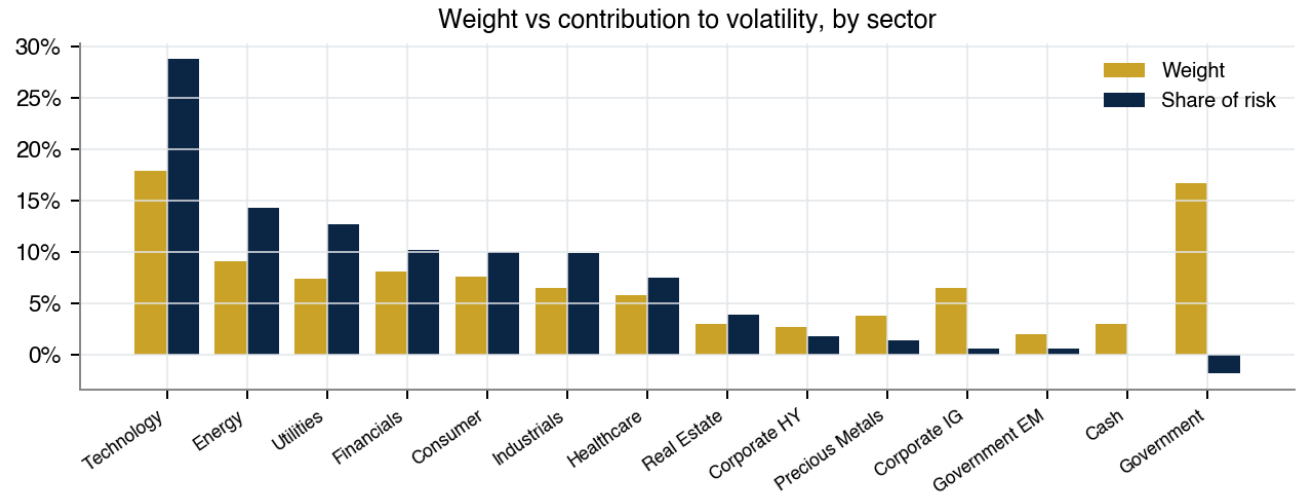


Figure 4. Weight vs share of risk. Source: tables/risk_contrib_sector.csv

Position	Weight	Stand-alone vol	Share of risk
Synthetic Equity 01	8.9%	27.6%	14.0%
Synthetic Equity 07	3.7%	32.5%	6.7%
Synthetic Equity 08	3.0%	39.4%	5.9%
Synthetic Equity 25	3.4%	30.2%	4.8%
Synthetic Equity 26	2.8%	35.4%	4.7%
Synthetic Equity 24	3.1%	31.7%	4.4%
Synthetic Equity 13	2.6%	34.2%	4.1%
Global REIT Index	3.0%	19.9%	3.9%
Synthetic Equity 12	1.9%	33.2%	3.7%
Synthetic Equity 28	2.2%	38.3%	3.5%
Synthetic Equity 02	2.1%	25.1%	3.2%
Synthetic Equity 19	2.4%	27.6%	3.2%

Source: tables/risk_contributions.csv

Asset class	Weight	Share of risk
Equity	62.4%	93.5%
Real Estate	3.0%	3.9%
Commodities	3.8%	1.4%
Fixed Income	27.9%	1.2%
Cash	3.0%	0.0%

Source: tables/risk_contrib_asset_class.csv

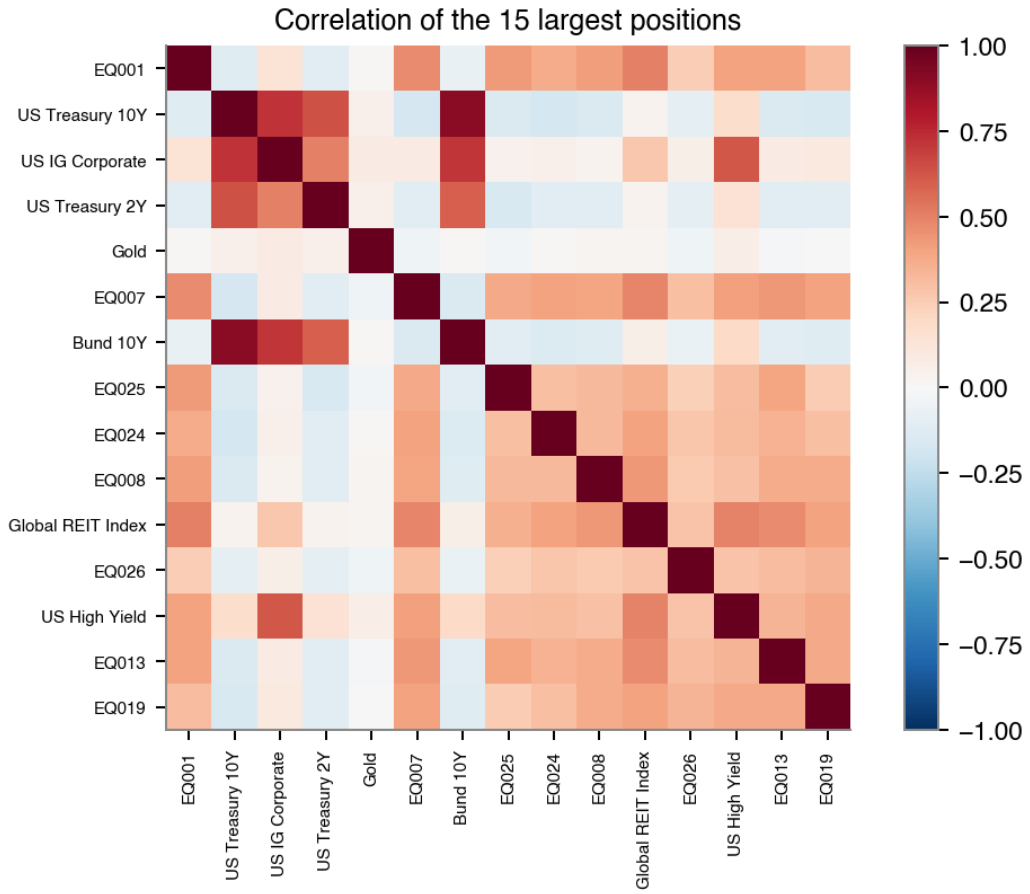


Figure 5. Correlation matrix. Source: tables/correlation_top15.csv

4. Factor exposure

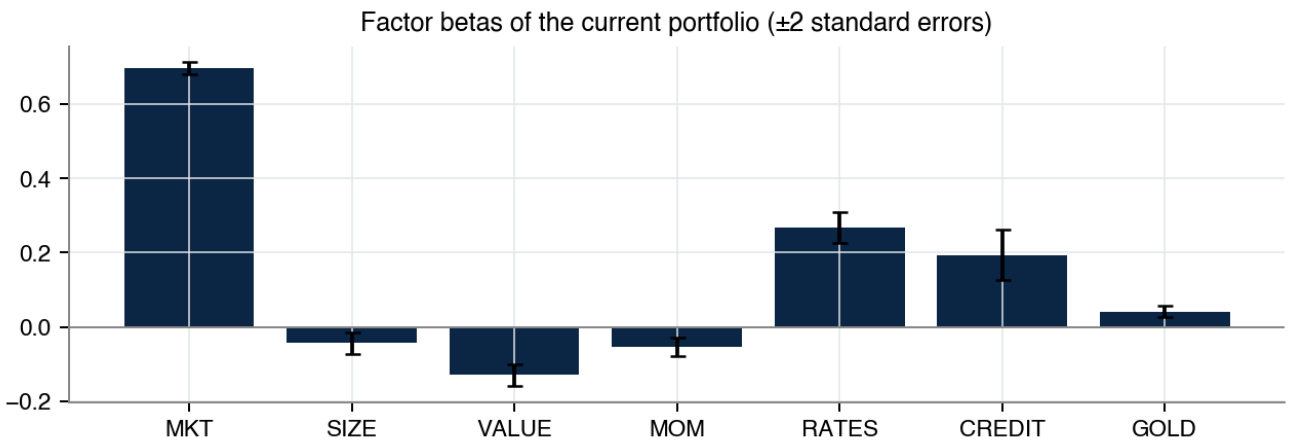


Figure 6. Factor betas. Source: tables/factor_regression.csv

Factor	Beta	t-stat
alpha	0.00	0.22
MKT	0.70	82.37
SIZE	-0.04	-2.97
VALUE	-0.13	-8.84
MOM	-0.05	-4.39
RATES	0.27	12.74
CREDIT	0.19	5.61
GOLD	0.04	5.16

Source: tables/factor_regression.csv

R^2 0.93: 79.4% of the portfolio's variance is explained by the factors; residual (stock-specific) volatility 3.3% a year.

5. Value-at-Risk and Expected Shortfall

Method	VaR 95%	ES 95%	VaR 97.5%	ES 97.5%	VaR 99%	ES 99%
Historical simulation	1.17%	1.72%	1.55%	2.16%	2.01%	2.78%
Parametric (normal)	1.41%	1.77%	1.68%	2.01%	2.00%	2.29%
Parametric (Cornish-Fisher)	1.24%	n/a	2.00%	n/a	3.22%	n/a
Monte Carlo	1.32%	1.91%	1.70%	2.32%	2.25%	2.90%

Source: tables/var_es.csv (1-day, % of NAV)

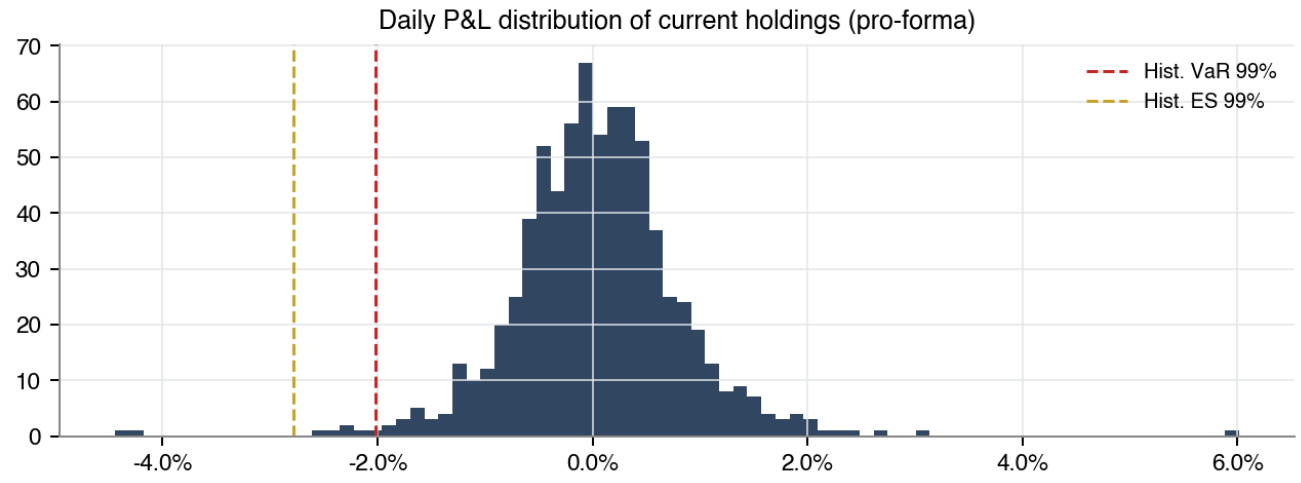


Figure 7. Pro-forma daily P&L distribution. Source: tables/pro_forma_returns.csv

Model backtest

5 exceptions in 500 days at 99% confidence (5.0 expected). Kupiec p-value 1.00; Christoffersen independence p-value 0.75; Basel zone **GREEN**.

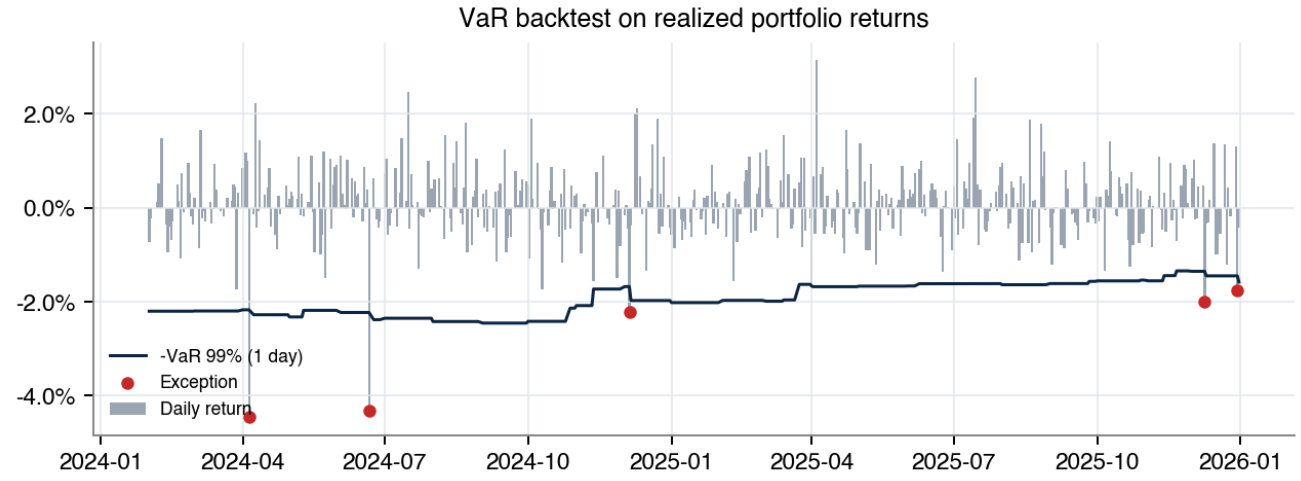


Figure 8. Realized returns vs VaR. Source: tables/var_backtest.csv

6. Stress tests

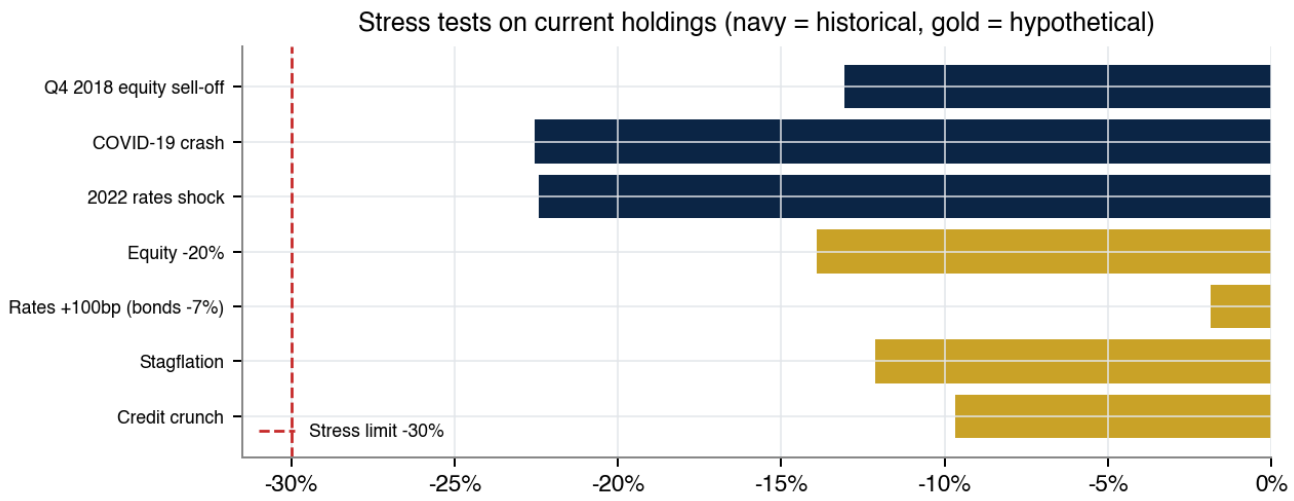


Figure 9. Scenario losses on current holdings. Source: tables/stress_tests.csv

Scenario	Type	Portfolio	Benchmark	P&L
Q4 2018 equity sell-off	historical	-13.1%	-10.6%	-USD 47.4m
COVID-19 crash	historical	-22.6%	-21.6%	-USD 81.7m
2022 rates shock	historical	-22.4%	-19.2%	-USD 81.3m
Equity -20%	hypothetical	-13.9%	n/a	-USD 50.4m
Rates +100bp (bonds -7%)	hypothetical	-1.9%	n/a	-USD 6.8m
Stagflation	hypothetical	-12.1%	n/a	-USD 43.9m
Credit crunch	hypothetical	-9.7%	n/a	-USD 35.1m

Source: tables/stress_tests.csv

Largest contributors in the worst historical scenario

Position	Sector	Contribution
Synthetic Equity 01	Technology	-1.90%
Synthetic Equity 07	Utilities	-1.63%
Synthetic Equity 26	Energy	-1.20%
Synthetic Equity 08	Technology	-1.19%
Synthetic Equity 25	Industrials	-1.17%
Synthetic Equity 29	Technology	-0.92%
Synthetic Equity 19	Energy	-0.90%
Synthetic Equity 28	Utilities	-0.89%
Global REIT Index	Real Estate	-0.86%
Synthetic Equity 05	Energy	-0.85%

Source: tables/stress_worst_contributors.csv

7. Monthly monitoring

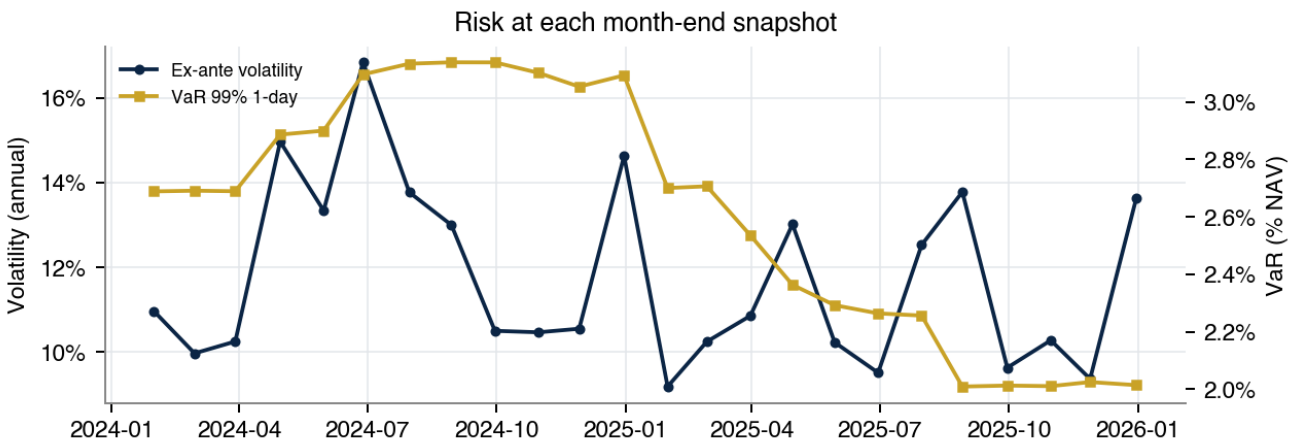


Figure 10. Risk at each month-end. Source: tables/monitoring_history.csv

Change vs previous month

Metric	Previous	Current	Change
Net asset value	USD 371.2m	USD 362.3m	-USD 8.9m
Ex-ante volatility	9.4%	13.6%	4.3%
VaR 99% 1-day	2.03%	2.01%	-0.01%
ES 97.5% 1-day	2.16%	2.16%	-0.00%
Top-10 weight	49.2%	49.1%	-0.1%
Largest position	9.3%	8.9%	-0.4%
Effective positions	24.54	24.73	0.19
Equity beta	0.71	0.70	-0.01
Positions	39	39	0

Source: tables/monitoring_changes.csv

Largest position changes

Position	Previous	Current	Change
US Treasury 10Y	8.42%	8.61%	0.20%
Synthetic Equity 25	3.26%	3.43%	0.17%
Synthetic Equity 27	0.88%	1.02%	0.14%
Synthetic Equity 08	2.89%	3.02%	0.14%
Synthetic Equity 28	2.02%	2.15%	0.13%
Synthetic Equity 01	9.31%	8.87%	-0.44%
Synthetic Equity 29	2.20%	1.88%	-0.33%
Synthetic Equity 24	3.23%	3.07%	-0.16%
Synthetic Equity 13	2.74%	2.59%	-0.15%
Bund 10Y	3.63%	3.54%	-0.09%

Source: tables/position_changes.csv

Limitations and assumptions

- Risk is estimated from historical returns; future volatility and correlations can differ, particularly in regimes not present in the look-back window.
- Square-root-of-time scaling assumes independent daily returns; autocorrelation or volatility clustering make multi-day risk differ from scaled one-day figures.
- Positions with short histories rely on peer proxies (table below).
- Hypothetical stress tests are linear in factor betas and ignore convexity, liquidity and second-round effects.
- Liquidity risk (time to liquidate positions) is not measured unless trading volumes are provided.
- Holdings and prices are as provided by the client / administrator; we reconciled internal consistency only.
- This demonstration uses synthetic data; its numbers have no meaning for any real portfolio.

Instrument	Weight	Proxied days (lookback)
EQ031	1.19%	598

Source: tables/proxy_usage.csv

Appendix

A.1 Configuration used

As-of date	2025-12-31
Look-back	750 days; covariance ewma (λ 0.94)
Confidence levels / horizons	[0.95, 0.975, 0.99] / [1, 10] days
Monte Carlo	{'n_sims': 20000, 'dist': 't', 'df': 5, 'seed': 7}
Factors	MKT, SIZE, VALUE, MOM, RATES, CREDIT, GOLD
Historical scenarios	Q4 2018 equity sell-off (2018-10-01 to 2018-12-24); COVID-19 crash (2020-02-20 to 2020-03-23); 2022 rates shock (2022-01-03 to 2022-10-12)

Source: config_used.yaml in the run folder

A.2 Data cleaning log

step	action	rows_affected	key
holdings	drop exact duplicate lines (confirm with administrator)	1	
holdings	snapshots	24	
prices	forward-filled interior gaps (max 3 days)	8	
prices	stale price run flagged (kept)	6	EQ012
proxies	returns proxied by asset_class+sector peers	2456	EQ031

Source: tables/cleaning_log.csv

Data quality

benchmark · 2609 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

factors · 2609 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

holdings · 943 rows · PASSED (0 errors, 2 warnings)

Check	Severity	Column	Count	Message
duplicate_rows	warning		1	1 fully duplicated rows
duplicate_keys	warning		2	2 rows share a key ['as_of_date', 'instrument_id'] (0 with conflicting values)

Source: data_quality/dq_holdings.json

holdings_clean · 942 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

prices · 99295 rows · PASSED (0 errors, 2 warnings)

Check	Severity	Column	Count	Message
missing_values	warning	close	8	8 missing (0.01%)
outliers	warning	close (daily return)	36	36 daily returns with robust |z| > 10.0

Source: data_quality/dq_prices.json

Audit trail

Run ID	`pr_20261111T142007Z_5ae784`
Run started / finished (UTC)	2026-11-11T14:20:07+00:00
Analyst	info@proquantanalytics.com
Code version (git)	`3a6bb15275c5 (uncommitted changes) · core 0.1.0`
Config SHA-256	`32c446c361b462e3`
Input: holdings	`943 rows · sha256 7cc200fef0cfd107`
Input: prices	`99295 rows · sha256 efd3f25952de6bb3`
Input: benchmark	`2609 rows · sha256 5639df02765c8ca5`
Input: factors	`2609 rows · sha256 0e8b02ed5d06d1d4`
Environment	`python 3.13.5 · pandas 2.2.3 · numpy 2.1.3`

Source: audit.json in the run folder

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